

Ordinary Differential Equations

Exam 1 — Comprehensive Study Guide

Chapters 1–3 • First-Order Methods, Qualitative Analysis & Second-Order Linear Theory

How to use this guide. Work through it in four passes: (1) lock in the **vocabulary** so you can classify any equation on sight; (2) memorize the **algorithms & formula sheet**; (3) re-derive the four **worked masters** from a blank page until they are automatic; (4) attempt the **8 practice problems** cold, then grade yourself against the detailed key. Every algebraic transition below is written out in full — no skipped steps.

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1. Core Vocabulary & Concept Glossary

Order of a differential equation

The **order** is the highest derivative that appears. $y' + 3y = t$ is *first order*; $y'' + 4y' + 13y = 0$ is *second order*; $y^{(4)} - y = 0$ is *fourth order*.

Linear vs. Non-linear

An n th-order ODE is **linear** if it can be written

$$a_n(t)y^{(n)} + \cdots + a_1(t)y' + a_0(t)y = g(t),$$

i.e. the unknown y and *all* of its derivatives appear only to the first power, never multiplied together, and never inside a nonlinear function. The coefficients $a_k(t)$ and the forcing $g(t)$ may depend on t in any way.

An equation is **non-linear** if y or a derivative appears as y^2 , yy' , $\sin y$, e^y , $\sqrt{y'}$, etc.

- Linear: $t^2 y'' + e^t y' + (\cos t) y = t^3$ (variable coefficients are fine).
- Non-linear: $y'' + \sin y = 0$ (pendulum), $y' = y^2$, $yy'' = 1$.

Autonomous equation

A first-order ODE is **autonomous** if the independent variable does *not* appear explicitly on the right-hand side: $\frac{dy}{dt} = f(y)$. The slope depends only on the state y , never on time t . These are the equations analyzed with **phase lines**. Example: $y' = y(2 - y)$ is autonomous; $y' = ty$ is not.

Homogeneous (linear ODE sense)

A linear ODE is **homogeneous** if the forcing term is zero:

$$a_n(t)y^{(n)} + \cdots + a_0(t)y = 0.$$

If the right side is a nonzero $g(t)$, the equation is **nonhomogeneous**.

Warning — two different uses of “homogeneous.” In Chapter 1, some texts call $y' = F(y/x)$ a “homogeneous equation” (a substitution method). For Exam 1 second-order theory, “homogeneous” means *the RHS is 0*. Read from context.

Wronskian

For two functions y_1, y_2 the **Wronskian** is the determinant

$$W(y_1, y_2)(t) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'.$$

If $W(y_1, y_2)(t_0) \neq 0$ at some point, then y_1, y_2 are **linearly independent** and form a fundamental set. If y_1, y_2 solve the same linear homogeneous ODE, **Abel’s Theorem** guarantees W is either identically zero or never zero on the interval.

Fundamental solution set

For a 2nd-order linear homogeneous ODE, a **fundamental solution set** is a pair $\{y_1, y_2\}$ of linearly independent solutions ($W \neq 0$). Every solution is then a unique linear combination, the **general solution**:

$$y(t) = c_1 y_1(t) + c_2 y_2(t).$$

Transient vs. Steady-State

In a solution split as $y(t) = y_{\text{trans}}(t) + y_{\text{ss}}(t)$:

- The **transient** part $\rightarrow 0$ as $t \rightarrow \infty$ (it carries the influence of the initial condition and dies out — typically the decaying exponential / homogeneous piece).
- The **steady-state** part is what remains as $t \rightarrow \infty$ (the long-run behavior — e.g. the limiting salt amount in a tank, or the particular response to forcing).

Example: in $Q(t) = 300 - 280e^{-t/50}$, the term $-280e^{-t/50}$ is transient and $Q_{ss} = 300$.

Damping classifications (mass–spring–damper)

For $my'' + \gamma y' + ky = 0$ the discriminant of the characteristic equation, $\gamma^2 - 4mk$, decides the behavior:

- **Undamped** ($\gamma = 0$): pure oscillation, $y = R \cos(\omega_0 t - \delta)$, $\omega_0 = \sqrt{k/m}$. Never decays.
- **Underdamped** ($\gamma^2 < 4mk$): complex roots $\lambda \pm i\mu$; *oscillation inside a decaying envelope* $e^{\lambda t}$. Crosses equilibrium infinitely often.
- **Critically damped** ($\gamma^2 = 4mk$): repeated real root; returns to equilibrium *fastest without oscillating*. Crosses equilibrium at most once.
- **Overdamped** ($\gamma^2 > 4mk$): two distinct negative real roots; *slow, non-oscillatory* return to equilibrium.

2. Detailed Methodology & Formula Sheet

2.1 Separable Equations

Algorithm — Separable: $\frac{dy}{dx} = g(x) h(y)$

1. **Separate** the variables: $\frac{dy}{h(y)} = g(x) dx$ (all y on the left, all x on the right).
2. **Integrate** both sides: $\int \frac{dy}{h(y)} = \int g(x) dx + C$. (One constant C is enough.)
3. **Solve for y** explicitly when possible; otherwise leave an implicit solution.
4. **Apply the initial condition** to find C ; **choose the correct branch** (sign of a square root) using the IC.
5. **Recover lost solutions:** any constant $y = r$ with $h(r) = 0$ is also an equilibrium solution that division may have discarded.

2.2 Linear First-Order via Integrating Factor

Algorithm — Integrating Factor

1. **Standard form:** write the equation as $y' + p(t)y = g(t)$ (leading coefficient = 1; divide if needed).
2. **Compute the integrating factor** $\mu(t) = e^{\int p(t) dt}$ (no $+C$ needed here).
3. **Multiply** the whole equation by μ . The left side collapses by the product rule into $(\mu(t)y)'$.
4. **Integrate:** $\mu(t)y = \int \mu(t)g(t) dt + C$.
5. **Solve for y** and apply the IC for C .

Why it works — the μ derivation

We want μ so that $\mu y' + \mu p y = (\mu y)' = \mu y' + \mu' y$. Matching the y -terms forces

$$\mu' = p\mu \implies \frac{\mu'}{\mu} = p \implies \ln |\mu| = \int p dt \implies \boxed{\mu(t) = e^{\int p(t) dt}}.$$

2.3 Modeling Templates**Mixing / Tank, Cooling, Free Fall**

Mixing tank (let $Q(t)$ = amount of substance, $V(t)$ = volume):

$$\frac{dQ}{dt} = \underbrace{(\text{conc. in})(\text{rate in})}_{\text{rate in}} - \underbrace{\frac{Q}{V(t)}(\text{rate out})}_{\text{rate out}}, \quad V(t) = V_0 + (r_{\text{in}} - r_{\text{out}})t.$$

Newton's Law of Cooling (T_m = ambient temperature):

$$\frac{dT}{dt} = -k(T - T_m), \quad T(t) = T_m + (T_0 - T_m)e^{-kt}, \quad k > 0.$$

Free fall with linear air resistance (downward positive, drag $\propto v$):

$$m \frac{dv}{dt} = mg - \gamma v, \quad v(t) = \underbrace{\frac{mg}{\gamma}}_{\text{terminal } v_\infty} + \left(v_0 - \frac{mg}{\gamma}\right)e^{-\gamma t/m}.$$

2.4 Euler's Method (Numerical Approximation)**Algorithm — Euler's Method for $y' = f(t, y)$, $y(t_0) = y_0$**

Pick a step size h . Then march forward using the tangent-line update:

$$t_{n+1} = t_n + h, \quad \boxed{y_{n+1} = y_n + h f(t_n, y_n)}.$$

Each step follows the slope field's slope *at the current point* for one step of width h . Smaller $h \Rightarrow$ smaller (local) error, more steps.

2.5 Second-Order Constant Coefficients — Characteristic Equation**The three root cases for $ay'' + by' + cy = 0$**

Substitute $y = e^{rt}$ to obtain the **characteristic equation** $ar^2 + br + c = 0$, with $r = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Discriminant	Roots	General solution
$b^2 - 4ac > 0$ (distinct real)	$r_1 \neq r_2$ (real)	$y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$
$b^2 - 4ac = 0$ (repeated real)	$r_1 = r_2 = r$	$y = (c_1 + c_2 t) e^{rt}$
$b^2 - 4ac < 0$ (complex)	$r = \lambda \pm i\mu$	$y = e^{\lambda t} (c_1 \cos \mu t + c_2 \sin \mu t)$

Euler's formula (source of the complex case): $e^{i\theta} = \cos \theta + i \sin \theta$, so $e^{(\lambda+i\mu)t} = e^{\lambda t}(\cos \mu t + i \sin \mu t)$; real and imaginary parts give the two real solutions.

2.6 Mass–Spring–Damper Parameters

Model: $m y'' + \gamma y' + k y = F(t)$

Symbol	Name	Meaning / how to get it
m	mass (inertia)	weight/ g , e.g. $m = W/g$.
γ	damping constant	drag/dashpot coefficient; $F_{\text{damp}} = -\gamma y'$.
k	spring constant	Hooke's law $k = \frac{mg}{L}$ from static stretch L ($mg = kL$).
ω_0	natural frequency	$\omega_0 = \sqrt{k/m}$ (undamped).
μ	quasi-frequency	imaginary part of the roots when underdamped.

Damping test: compute $\gamma^2 - 4mk$. < 0 underdamped, $= 0$ critically damped, > 0 overdamped.

3. Step-by-Step Worked Examples

Master Example 1 — Integrating-Factor IVP

Solve $t y' + 2y = t^2 - t + 1$, $y(1) = \frac{1}{2}$, $t > 0$.

Step 1 — Standard form. Divide every term by t (legal since $t > 0$):

$$y' + \frac{2}{t} y = t - 1 + \frac{1}{t}, \quad \text{so} \quad p(t) = \frac{2}{t}.$$

Step 2 — Integrating factor.

$$\mu(t) = e^{\int \frac{2}{t} dt} = e^{2 \ln t} = e^{\ln t^2} = t^2.$$

Step 3 — Multiply and collapse the left side.

$$t^2 y' + 2t y = t^3 - t^2 + t \implies \frac{d}{dt}(t^2 y) = t^3 - t^2 + t.$$

Check the collapse: $\frac{d}{dt}(t^2 y) = t^2 y' + 2t y$. ✓ It matches the left side exactly — that is the whole point of μ .

Step 4 — Integrate both sides.

$$t^2 y = \int (t^3 - t^2 + t) dt = \frac{t^4}{4} - \frac{t^3}{3} + \frac{t^2}{2} + C.$$

Step 5 — Solve for y . Divide by t^2 :

$$y(t) = \frac{t^2}{4} - \frac{t}{3} + \frac{1}{2} + \frac{C}{t^2}.$$

Step 6 — Apply $y(1) = \frac{1}{2}$.

$$\frac{1}{2} = \frac{1}{4} - \frac{1}{3} + \frac{1}{2} + C \implies 0 = \frac{1}{4} - \frac{1}{3} + C \implies C = \frac{1}{3} - \frac{1}{4} = \frac{4-3}{12} = \frac{1}{12}.$$

$$y(t) = \frac{t^2}{4} - \frac{t}{3} + \frac{1}{2} + \frac{1}{12t^2}$$

Master Example 2 — Mixture / Tank Word Problem

A tank holds 100 gallons of brine containing 20 lb of dissolved salt. Brine with concentration 3 lb of salt per gallon flows in at 2 gal/min; the well-stirred mixture flows out at the same 2 gal/min.

Find $Q(t)$, the amount of salt at time t , and the limiting amount.

Step 1 — Track the volume. Inflow rate = outflow rate = 2 gal/min, so the volume is constant:

$$V(t) = 100 + (2 - 2)t = 100 \text{ gallons.}$$

Step 2 — Build the rate balance.

$$\frac{dQ}{dt} = (\text{rate in}) - (\text{rate out}).$$

- Rate in = $\left(3 \frac{\text{lb}}{\text{gal}}\right) \left(2 \frac{\text{gal}}{\text{min}}\right) = 6 \frac{\text{lb}}{\text{min}}.$
- Rate out = $\left(\frac{Q}{100} \frac{\text{lb}}{\text{gal}}\right) \left(2 \frac{\text{gal}}{\text{min}}\right) = \frac{Q}{50} \frac{\text{lb}}{\text{min}}.$

$$\boxed{\frac{dQ}{dt} = 6 - \frac{Q}{50}}, \quad Q(0) = 20.$$

Step 3 — Put in standard linear form and find μ .

$$Q' + \frac{1}{50} Q = 6, \quad \mu(t) = e^{\int \frac{1}{50} dt} = e^{t/50}.$$

Step 4 — Multiply, collapse, integrate.

$$\frac{d}{dt}(e^{t/50} Q) = 6 e^{t/50} \implies e^{t/50} Q = 6 \int e^{t/50} dt = 6 \cdot 50 e^{t/50} + C = 300 e^{t/50} + C.$$

Step 5 — Solve for Q and apply $Q(0) = 20$.

$$Q(t) = 300 + C e^{-t/50}, \quad 20 = 300 + C \implies C = -280.$$

$$Q(t) = 300 - 280 e^{-t/50} \quad \text{and} \quad \lim_{t \rightarrow \infty} Q(t) = 300 \text{ lb.}$$

Sanity check (steady state). Set $Q' = 0$ in the ODE: $0 = 6 - \frac{Q}{50} \implies Q = 300$. The limiting salt equals (inflow concentration) $\times$ (volume) $= 3 \times 100 = 300$ lb. The transient piece $-280e^{-t/50}$ dies out. ✓

Master Example 3 — Autonomous Equation: Phase Line & Stability

Analyze $\frac{dy}{dt} = f(y) = y(y-2)(y+2) = y^3 - 4y$. Find all equilibria, classify their stability, and sketch the phase line.

Step 1 — Find equilibrium (critical) solutions. Set $f(y) = 0$:

$$y(y-2)(y+2) = 0 \implies y = -2, y = 0, y = 2.$$

Step 2 — Sign analysis of $f(y)$ on each interval. Test one point per interval:

Interval	Test y	$f(y)$	Flow of y
$y < -2$	$y = -3$	$(-3)(-5)(-1) = -15 < 0$	decreasing ($\leftarrow$)
$-2 < y < 0$	$y = -1$	$(-1)(-3)(1) = +3 > 0$	increasing ($\rightarrow$)
$0 < y < 2$	$y = 1$	$(1)(-1)(3) = -3 < 0$	decreasing ($\leftarrow$)
$y > 2$	$y = 3$	$(3)(1)(5) = +15 > 0$	increasing ($\rightarrow$)

Step 3 — Classify each equilibrium. An equilibrium is *stable* (sink) if arrows point toward it from both sides, *unstable* (source) if both point away.

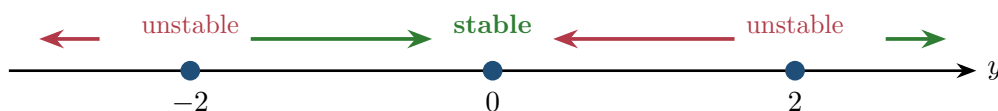
- $y = -2$: arrows $\leftarrow$ on the left, $\rightarrow$ on the right $\implies$ both *away* $\implies$ **unstable (source)**.
- $y = 0$: arrows $\rightarrow$ on the left, $\leftarrow$ on the right $\implies$ both *toward* $\implies$ **asymptotically stable (sink)**.
- $y = 2$: arrows $\leftarrow$ on the left, $\rightarrow$ on the right $\implies$ both *away* $\implies$ **unstable (source)**.

Step 4 — Confirm with the derivative test. $f'(y) = 3y^2 - 4$:

$$f'(-2) = 8 > 0 \text{ (unstable)}, \quad f'(0) = -4 < 0 \text{ (stable)}, \quad f'(2) = 8 > 0 \text{ (unstable)}.$$

Rule: $f'(y^*) < 0 \implies$ stable, $f'(y^*) > 0 \implies$ unstable. Both methods agree. ✓

Step 5 — Phase line.



Interpretation. Any initial value in $(-2, 2)$ is pulled toward $y = 0$. Start above 2 and $y \rightarrow +\infty$; start below -2 and $y \rightarrow -\infty$. The points $y = \pm 2$ are thresholds (separatrices).

Master Example 4 — 2nd-Order Constant Coefficients, Complex Roots

Solve the IVP $y'' + 4y' + 13y = 0$, $y(0) = 1$, $y'(0) = -1$.

Step 1 — Characteristic equation. Sub $y = e^{rt}$:

$$r^2 + 4r + 13 = 0.$$

Step 2 — Solve via the quadratic formula.

$$r = \frac{-4 \pm \sqrt{4^2 - 4(1)(13)}}{2(1)} = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = -2 \pm 3i.$$

So $\lambda = -2$ (real part) and $\mu = 3$ (imaginary part). Discriminant $< 0 \Rightarrow$ **complex case**.

Step 3 — Write the general solution.

$$y(t) = e^{\lambda t}(c_1 \cos \mu t + c_2 \sin \mu t) = e^{-2t}(c_1 \cos 3t + c_2 \sin 3t).$$

Step 4 — Apply $y(0) = 1$. At $t = 0$: $e^0(c_1 \cdot 1 + c_2 \cdot 0) = c_1$, so

$$c_1 = 1.$$

Step 5 — Differentiate (product rule) for $y'(0)$.

$$y'(t) = -2e^{-2t}(c_1 \cos 3t + c_2 \sin 3t) + e^{-2t}(-3c_1 \sin 3t + 3c_2 \cos 3t).$$

At $t = 0$:

$$y'(0) = -2c_1 + 3c_2 = -1.$$

Substituting $c_1 = 1$: $-2 + 3c_2 = -1 \Rightarrow 3c_2 = 1 \Rightarrow c_2 = \frac{1}{3}$.

$$y(t) = e^{-2t} \left(\cos 3t + \frac{1}{3} \sin 3t \right)$$

Step 6 — (Optional) Amplitude–phase form. With $R = \sqrt{c_1^2 + c_2^2} = \sqrt{1 + \frac{1}{9}} = \frac{\sqrt{10}}{3}$ and $\tan \delta = \frac{c_2}{c_1} = \frac{1}{3}$:

$$y(t) = \frac{\sqrt{10}}{3} e^{-2t} \cos(3t - \delta), \quad \delta = \arctan \frac{1}{3} \approx 0.3217 \text{ rad}.$$

This is an **underdamped** response: a 3-rad/s oscillation inside a decaying envelope $\pm \frac{\sqrt{10}}{3} e^{-2t}$.

4. Practice Problem Set

Attempt all eight *before* looking at the key. Time target: ~ 60 minutes. Full worked solutions begin on the next section.

1. **(Classification)** State the order, and whether each is linear or non-linear:

(a) $y'' + 3t y' + \sin(y) = e^t$

(b) $\frac{dy}{dx} = \frac{x^2 y}{1+x}$

2. **(Separable IVP)** Solve $\frac{dy}{dx} = \frac{3x^2 + 4x + 2}{2(y-1)}$, $y(0) = -1$. Give y explicitly.

3. **(Integrating Factor IVP)** Solve $y' + 2y = t e^{-2t}$, $y(1) = 0$.

4. **(Newton's Cooling)** A cup of coffee at 200°F sits in a 70°F room. After 5 minutes it has cooled to 160°F . Find $T(t)$, then determine when the coffee reaches 100°F .

5. **(Euler's Method)** For $\frac{dy}{dt} = t + 2y$, $y(0) = 1$, use step size $h = 0.1$ to approximate $y(0.3)$. Show the table.

6. **(Autonomous / Phase Line)** For $\frac{dy}{dt} = y^2 - y - 6$, find all equilibria and classify each as stable or unstable.

7. **(Repeated Roots IVP)** Solve $y'' + 6y' + 9y = 0$, $y(0) = 2$, $y'(0) = -1$.

8. **(Mechanical Vibrations)** A mass–spring–damper system satisfies $m = 1$, $\gamma = 2$, $k = 5$ (consistent units): $y'' + 2y' + 5y = 0$, with $y(0) = 1$, $y'(0) = 0$. Classify the damping and solve the IVP.

5. Answer Key & Detailed Solutions

Problem 1 — Classification

(a) Highest derivative is $y'' \Rightarrow$ **second order**. The term $\sin(y)$ is a nonlinear function of the unknown $\Rightarrow$ **non-linear**. (Note $3t y'$ alone is fine — variable coefficients are allowed — it is the $\sin y$ that breaks linearity.)

(b) Highest derivative is $\frac{dy}{dx} \Rightarrow$ **first order**. Rewriting, $y' - \frac{x^2}{1+x} y = 0$: the unknown y appears only to the first power with a coefficient in $x \Rightarrow$ **linear** (and also separable).

Problem 2 — Separable IVP

Separate and integrate:

$$2(y-1) dy = (3x^2 + 4x + 2) dx \implies \int 2(y-1) dy = \int (3x^2 + 4x + 2) dx.$$

$$y^2 - 2y = x^3 + 2x^2 + 2x + C.$$

Apply $y(0) = -1$: $(-1)^2 - 2(-1) = 1 + 2 = 3 = C$. So $y^2 - 2y = x^3 + 2x^2 + 2x + 3$. Complete the square on the left, $y^2 - 2y + 1 = (y-1)^2$:

$$(y-1)^2 = x^3 + 2x^2 + 2x + 4 \implies y-1 = \pm \sqrt{x^3 + 2x^2 + 2x + 4}.$$

At $x = 0$ we need $y = -1$, i.e. $y-1 = -2 < 0$, so take the *negative* branch:

$$\boxed{y(x) = 1 - \sqrt{x^3 + 2x^2 + 2x + 4}}.$$

Problem 3 — Integrating Factor IVP

Already in standard form with $p(t) = 2$, so $\mu = e^{\int 2 dt} = e^{2t}$. Multiply:

$$\frac{d}{dt}(e^{2t}y) = e^{2t} \cdot t e^{-2t} = t \implies e^{2t}y = \frac{t^2}{2} + C.$$

$$y = \left(\frac{t^2}{2} + C\right)e^{-2t}.$$

Apply $y(1) = 0$: $(\frac{1}{2} + C)e^{-2} = 0 \Rightarrow C = -\frac{1}{2}$.

$$y(t) = \frac{t^2 - 1}{2} e^{-2t}.$$

Problem 4 — Newton's Law of Cooling

Model: $T' = -k(T - 70)$, with $T_0 = 200$, $T_m = 70$:

$$T(t) = 70 + (200 - 70)e^{-kt} = 70 + 130e^{-kt}.$$

Use $T(5) = 160$: $160 = 70 + 130e^{-5k} \Rightarrow e^{-5k} = \frac{90}{130} = \frac{9}{13}$. Hence

$$k = \frac{1}{5} \ln \frac{13}{9} \approx 0.0735 \text{ min}^{-1}, \quad T(t) = 70 + 130 \left(\frac{9}{13}\right)^{t/5}.$$

Set $T = 100$: $100 = 70 + 130e^{-kt} \Rightarrow e^{-kt} = \frac{30}{130} = \frac{3}{13}$. Thus

$$t = \frac{\ln(13/3)}{k} = \frac{5 \ln(13/3)}{\ln(13/9)} \approx \frac{5(1.4663)}{0.3677} \approx \boxed{19.9 \text{ minutes}}.$$

Problem 5 — Euler's Method

$f(t, y) = t + 2y$, $h = 0.1$, $y_{n+1} = y_n + 0.1(t_n + 2y_n)$:

n	t_n	y_n	$y_{n+1} = y_n + 0.1(t_n + 2y_n)$
0	0.0	1.00	$1 + 0.1(0 + 2.00) = 1.20$
1	0.1	1.20	$1.20 + 0.1(0.1 + 2.40) = 1.45$
2	0.2	1.45	$1.45 + 0.1(0.2 + 2.90) = 1.76$
3	0.3	1.76	—

$$y(0.3) \approx 1.76.$$

Problem 6 — Autonomous / Phase Line

$f(y) = y^2 - y - 6 = (y - 3)(y + 2)$, so equilibria are $y = 3$ and $y = -2$. Derivative test $f'(y) = 2y - 1$:

$$f'(3) = 5 > 0 \Rightarrow \text{unstable}, \quad f'(-2) = -5 < 0 \Rightarrow \text{stable}.$$

(Sign check: $f < 0$ on $(-2, 3)$ so the flow there is toward -2 and away from 3 , confirming $y = -2$ is a sink and $y = 3$ is a source.)

$$y = -2 \text{ stable (sink); } y = 3 \text{ unstable (source)}.$$

Problem 7 — Repeated Roots IVP

Characteristic: $r^2 + 6r + 9 = (r + 3)^2 = 0 \Rightarrow r = -3$ (repeated). General solution:

$$y(t) = (c_1 + c_2 t) e^{-3t}.$$

$y(0) = c_1 = 2$. Differentiate: $y' = c_2 e^{-3t} - 3(c_1 + c_2 t) e^{-3t}$, so $y'(0) = c_2 - 3c_1 = -1 \Rightarrow c_2 = -1 + 6 = 5$.

$$\boxed{y(t) = (2 + 5t) e^{-3t}}.$$

Problem 8 — Mechanical Vibrations

Damping test: $\gamma^2 - 4mk = 2^2 - 4(1)(5) = 4 - 20 = -16 < 0 \Rightarrow$ **underdamped**.

Characteristic: $r^2 + 2r + 5 = 0 \Rightarrow r = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$, so $\lambda = -1$, $\mu = 2$:

$$y(t) = e^{-t}(c_1 \cos 2t + c_2 \sin 2t).$$

$y(0) = c_1 = 1$. Then

$$y'(t) = -e^{-t}(c_1 \cos 2t + c_2 \sin 2t) + e^{-t}(-2c_1 \sin 2t + 2c_2 \cos 2t),$$

$$y'(0) = -c_1 + 2c_2 = 0 \Rightarrow c_2 = \frac{c_1}{2} = \frac{1}{2}.$$

$$\boxed{y(t) = e^{-t} \left(\cos 2t + \frac{1}{2} \sin 2t \right)} \quad (\text{decaying oscillation; quasi-frequency } \mu = 2).$$

End of Study Guide — good luck on Exam 1.